Data Processing of EPMA and LA-ICP-MS analyses of basaltic glasses, olivines and melt inclusions (Set 4)

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1 Introduction

This report takes three raw instrument datasets, two electron microprobe (EPMA) sessions and one laser-ablation ICP-MS (LA-ICP-MS) session, from the exported files to corrected, quality-checked data, and gives the reason for every processing decision. The material is volcanic: basaltic glasses (81 spots on eight samples), olivine crystals (113 spots on 37 grains from three samples) and melt inclusions (47 analyses of 41 inclusions in four suites, MC22, MC59, MC82 and MC99U). Together they are the usual toolkit for following a magma: the glass records the melt at eruption, the olivine the crystallising phase, and the inclusions melt trapped earlier, before degassing.

The question addressed here is analytical rather than petrological. Both instruments already return matrix-corrected concentrations (CITZAF for the EPMA, GLITTER for the LA-ICP-MS), and the task is to establish how far they can be trusted: how reproducible each element is (precision), how close the standards come to their reference values (accuracy), whether the instrument response changed during the session (drift), which analyses missed the intended material, and which elements lie too close to the detection limit to be reported. The same logic is applied to the three sessions: a correction factor derived from a standard analysed with the samples, tested on standard analyses held out of the correction. The full data tables are in the Appendix.

2 Detailed methodology

2.1 Common approach

Each session starts with the standards. Every element of a standard is plotted against the order of analysis, and a value is removed only when it is clearly separated from the other analyses of that standard and pulls the mean; a shift shared by a whole standard bracket is kept, because it also affected the samples measured at the same time. The mean of the kept analyses gives CF = mean/reference, and every analysis, samples and standards, is divided by its CF: the standard was measured under the same conditions as the samples, so its bias is assumed to be theirs too. CF values more than 5 % from 1 are flagged, as they point to a calibration problem. Because the corrected standard equals its reference by construction, accuracy is estimated by recomputing the CF without some of the standard analyses and applying it to those held-out analyses, which then behave as unknowns. Precision is given as 2RSD (2σ). Time-intervals are assumed to be constant across measurements therefore the analysis number was used as time when needed.

2.2 EPMA, olivine (steps 1 to 10)

Steps 1–2. The oxide (wt%) and Fo/Fa (mol%) files were joined on the analysis number, giving 128 analyses: 15 of SC-Olivine (five brackets of three) and 113 spots on 37 grains. **Step 3.** Three SC-Olivine values were removed (Fig. 1a): NiO of No. 90 (0.283 against 0.34–0.38 wt%), Cr_2O_3 of No. 1 (0.031 against 0.005–0.022 wt%) and Al_2O_3 of No. 128 (0.044 against 0.029–0.039 wt%). SiO_2 rises from 39.5–39.8 wt% in the first three brackets to 41.0–41.3 wt% in the last; as a trend, not a single bad point, it was kept. **Step 4.** Reference values come from `Standards.xlsx`, except SiO_2, MgO and FeO, taken from `_std-cnd`: the same values with more digits, and exactly those the instrument was calibrated on. **Step 5.** Only CaO falls outside ± 5 % (CF = 1.106; Fig. 2a). **Step 6.** The CF was recomputed from the first two analyses of each bracket (n = 10) and applied to the third (n = 5), so that the test analyses span the whole session. **Step 7.** Fo in the ratios file is the molar ratio $100 \times \mathrm{Mg}/(\mathrm{Mg}+\mathrm{Fe})$, with $\mathrm{Mg} = \mathrm{MgO}/40.30$ and $\mathrm{Fe} = \mathrm{FeO}/71.84$, calculated by the microprobe software from the same MgO and FeO (reproduced here within 0.009 mol%), so it is not an independent measurement and cannot recalibrate MgO or FeO. The correction changes Fo by -0.14 mol% on average; SC-Olivine gives Fo 90.23 ± 0.12 against 90.22. **Step 8.** Precision and accuracy of the corrected standard (Section 3). **Step 9.** Totals outside 98–102 wt% were flagged; olivine contains no volatiles, so a low total would not be admissible. No analysis is obviously wrong for its grain: Al_2O_3 ($\leq$0.05 wt%) and CaO ($\leq$0.37 wt%) show no beam overlap onto glass or inclusions. Values below the detection limit of their own analysis (from `_all`) were flagged, never removed. **Step 10.** Grain means (n, mean, 2SD).

2.3 EPMA, glass

The glass follows the same ten steps; only those that differ are given here (step 7, on Fo, applies to olivine only). **Standard.** Only VG-2 is used. This is our choice; the requirement is that the standard has a reference value for every oxide to be corrected, and VG-2 is the only standard of the session with all 13 (VGA99 has 12, scapolite 9, KE3 3). It is also a basaltic glass like the samples; the instrument calibration on VGA99, scapolite, KE3 and rhodonite stays underneath this correction. **Step 3.** SiO_2, Al_2O_3 and Cl of analysis No. 100 were removed (50.39 against 51.17–51.66 wt% SiO_2, 13.55 against 13.96–14.07 wt% Al_2O_3, 0.022 against 0.030–0.035 wt% Cl; Fig. 1b). No. 100 also has the lowest total (100.0 against 101.0–101.7 wt%), while No. 101, measured right after it, is normal: a poor analysis, not a change of the instrument. **Step 4.** Reference values come from `Standards.xlsx`, with S converted to SO_3. **Step 5.** MgO (1.061), P_2O_5 (0.944) and F (1.491) fall outside ± 5 % (Fig. 2b). **Step 6.** The CF was computed from analyses No. 3, 4, 57 and 100 and tested on No. 56 and 101. **Step 9.** A total below 98 wt% can be admissible for a glass, which may contain volatiles that are not measured. Two analyses are ob-

Table 1: Analytical sessions and standards, from the `_std-cnd` and `_all` files and the GLITTER export. Numbers in parentheses are numbers of analyses, unless stated otherwise.

	EPMA glass	EPMA olivine	LA-ICP-MS
Session	20 Dec 2016	2 Mar 2017	19 Sep 2017
Conditions	JEOL WDS, 15 kV, 10 nA, 5 µm beam; 10–30 s peak, 10–15 s background	JEOL WDS, 15 kV, 100 nA, focused beam; 20–60 s peak, 10–30 s background	GLITTER 4.4.4; internal standard ^{43}Ca
Instrument calibration	VGA99 (9 oxides), scapolite (SO_3, Cl), KE3 (F), rhodonite (MnO)	SC-Olivine (SiO_2, MgO, FeO), wollastonite, rhodonite, corundum, Cr_2O_3, NiO	KL2-G, bracketing every ∼20 analyses
Correction (this study)	VG-2, one CF per oxide (5–6)	SC-Olivine, one CF per oxide (14–15)	KL2-G train split, CF fitted against time (11)
Accuracy test	VG-2 held out (2)	SC-Olivine held out (5)	KL2-G test split (5); BCR-2G (3), GOR128-G, GOR132-G, BM90/21-G
Standards / samples (n)	24 / 81	15 / 113	38 / 47

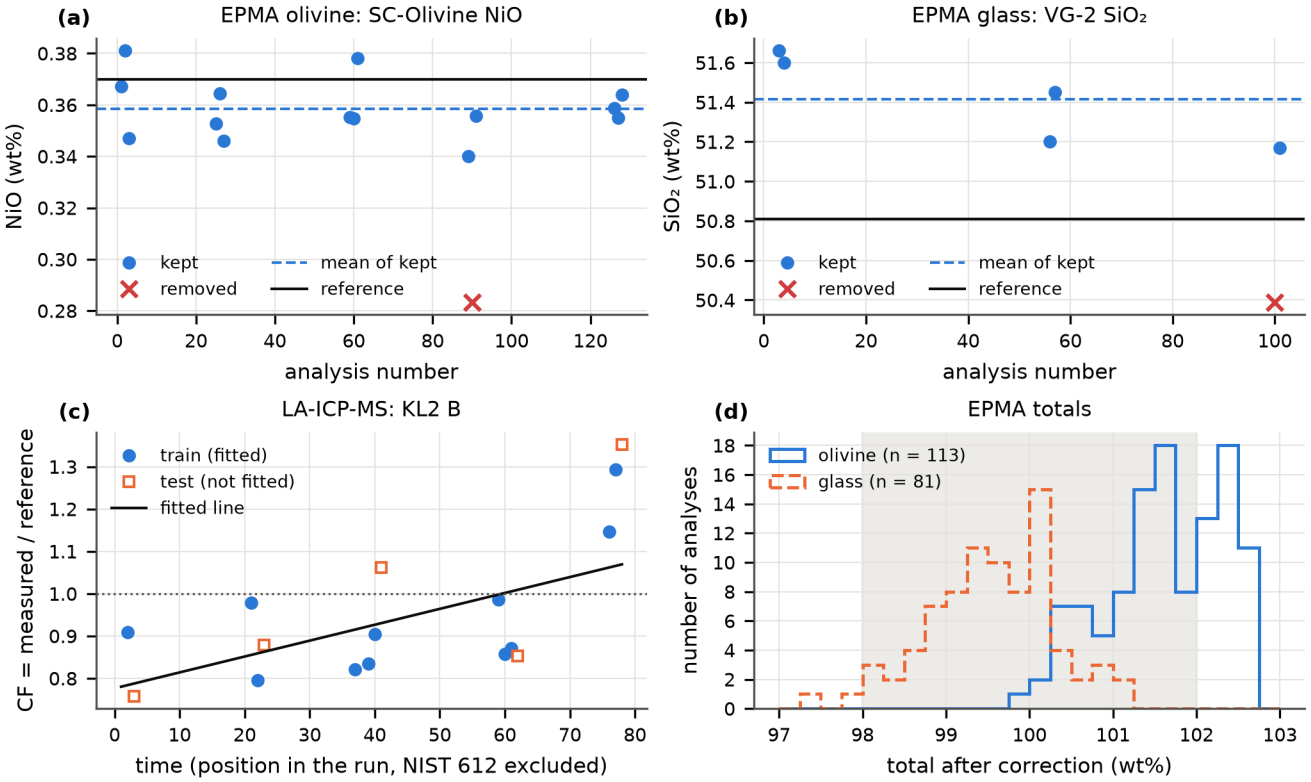


Figure 1: Examples of the processing steps. (a) SC-Olivine NiO and (b) VG-2 SiO_2 against analysis number, with the value removed in step 3, the mean of the kept analyses (dashed) and the reference value (solid). (c) Correction factor of KL2-G boron against time: the train analyses define the line used to correct every analysis, the test analyses are not fitted; boron has the strongest trend of all elements. (d) Analytical totals after correction, with the 98–102 wt% window (grey).

viously not glass of their sample and were excluded: No. 49 has the composition of clinopyroxene (13.8 wt% MgO, 23.3 wt% CaO, 6.4 wt% Al_2O_3) and No. 65 has 8.6 wt% Na_2O (the highest raw total, 102.5 wt%). Two single values were excluded as well, SO_3 of No. 32 (2.2 against 0.08–0.13 wt%) and Cl of No. 70 (0.70 against 0.02–0.22 wt%). Flow1-1, measured twice under one name (No. 71–79 and 89–97), was kept as two groups, which agree within 2SD.

2.4 LA-ICP-MS (steps 1 to 7)

Step 1. The GLITTER export (one column per analysis, several stacked blocks) was transposed to one row per analysis: 85 analyses of 41 isotopes. Reference values come from `Standards.xlsx` (GeoReM). **Step 2.** The MDL-filtered block, in which GLITTER replaces values below the minimum detection limit by text, is used to build the correction; the correction is then applied to the block not filtered for MDL, so that no value is lost, and values below the MDL are flagged. NIST 612 (runs 1–3, 27, 84–86) only monitors the machine, so it is ignored and not counted as time. Time is the position of each remaining analysis in the run, assuming equal intervals (004KL2 = 1, 083KL2 = 78). KL2-G was split into a train set (runs 5, 24, 25, 42, 44, 45, 64, 65, 66, 81, 82), a test set (runs 6, 26, 46, 67, 83) and one discarded analysis (run 4). **Step 3.** For each train analysis and element, CF = measured/reference (KL2-G values in `standard_icpms.csv`, identical to GeoReM), and a straight line was fitted to CF against time (Fig. 1c). No outliers were removed. ^{43}Ca is not corrected: as the internal standard it is fixed in GLITTER (77 901.7 µg/g in every KL2-G analysis, so its CF is 1.000). **Step 4.** Every analysis was

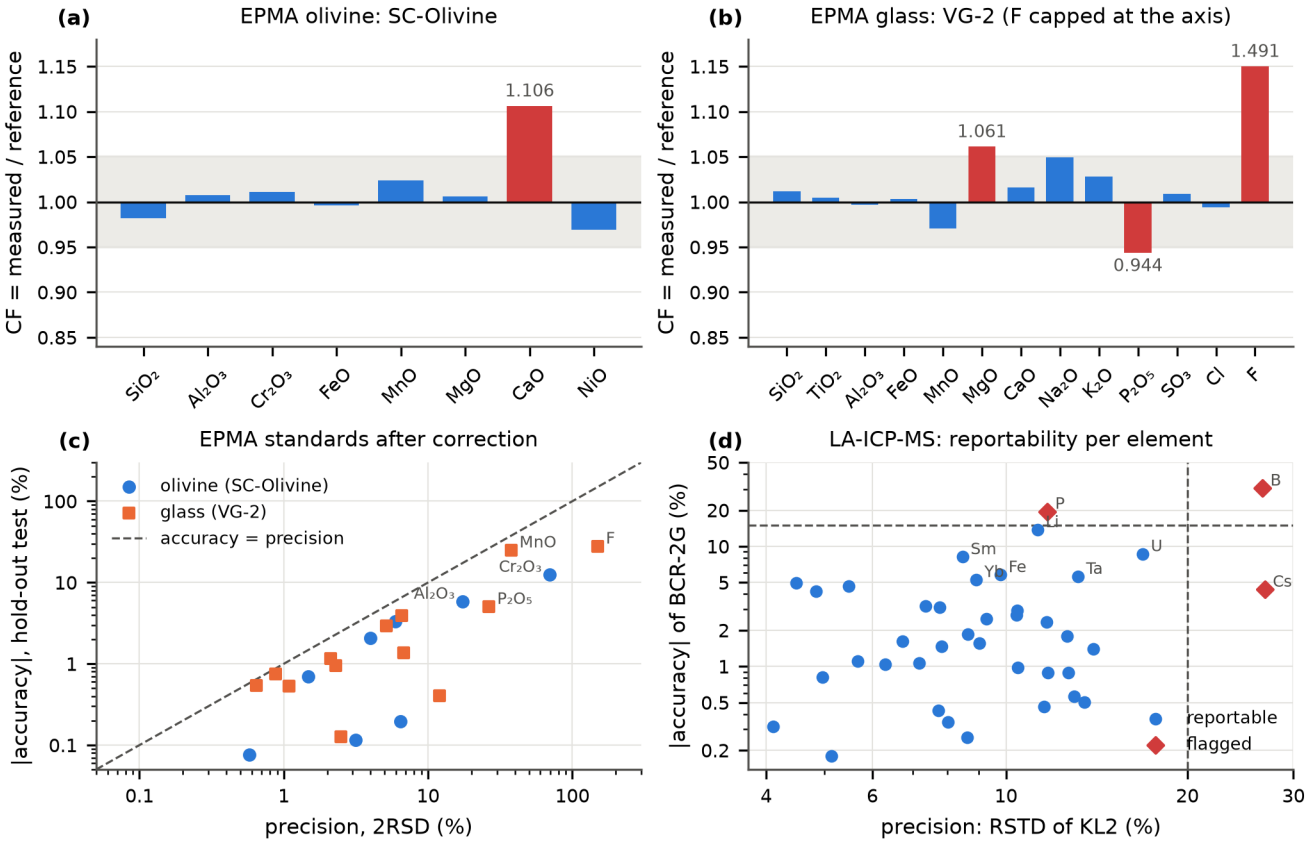


Figure 2: Data quality. (a, b) Correction factors per oxide for the olivine and glass sessions; red: outside ±5 % (grey band), with values (F is cut at the axis). (c) Precision (2RSD of the corrected standard) against accuracy (held-out standard analyses) in both EPMA sessions; labelled: |accuracy| ≥ 5 %. (d) LA-ICP-MS: precision (RSTD of KL2-G, train and test) against accuracy (BCR-2G); flagged elements in red, thresholds dashed. Ca (internal standard) and As (no BCR-2G reference) are not shown.

corrected as $c = c_{\mathrm{raw}}/(at+b)$, with slope a, intercept b and time t. Dividing, as for the EPMA, removes the deviation; multiplying by a factor defined as measured/reference would add it a second time (on the train split, the median deviation of KL2-G is 0.84 % uncorrected, 0.00 % after division and 1.67 % after multiplication). **Step 5.** Average, STDev, 2STDev, RSTD (2STDev/average) and accuracy were computed for KL2-G (train, test and both) and for BCR-2G, GOR132-G, GOR128-G and BM90/21-G (Table A4). **Step 6.** The 12 values below the MDL (As, Nb, Ta, Th and U, all in the depleted GOR and BM90 glasses; none in KL2-G or the inclusions) were excluded. BCR-2G run 70 was excluded: GLITTER normalised it to 71 470 µg/g Ca instead of the 50 900 used for the other BCR-2G analyses, so every element reads 0.55 to 5.6 times its reference (median 1.74). Elements were flagged when the RSTD of KL2-G exceeds 20 % or the BCR-2G accuracy is worse than ±15 %. **Step 7.** REE were normalised to CI chondrite and the multi-element pattern to primitive mantle, both from Sun and McDonough (1989): the most widely used values for basalts, and from one source so that the two patterns are consistent. Elements with too much range were reported (Pb, Section 3), and the inclusions were checked for host olivine with their major-element oxide sums.

3 Assessment of data quality

Precision. Major oxides reproduce at 0.6–3.1 % (2RSD; Fig. 2c, Tables A1–A2), best for MgO in SC-Olivine and Al_2O_3 in VG-2 (0.6 %). Minor elements are less precise, depending on concentration, count rate and counting time: Na_2O, K_2O and SO_3 5–7 %, Cl 12 %, P_2O_5 26 %, MnO 38 % and F 150 % in the glass; MnO 5.9 %, NiO 6.5 %, Al_2O_3 17 % and Cr_2O_3 70 % in the olivine. The higher current and longer counting of the olivine session (100 nA and 20–60 s, against 10 nA and 10–30 s) explain why its minor elements are better (MnO 5.9 against 38 %). In the LA-ICP-MS session KL2-G reproduces at 4–17 % RSTD (median 9 %) for the reported elements.

Accuracy. Every EPMA oxide plots below the accuracy = precision line (Fig. 2c): after correction, no oxide deviates from its reference by more than its own scatter, the criterion for a problem. The held-out analyses agree within 1.2 % for SiO_2, TiO_2, Al_2O_3, FeO, MgO and CaO in the glass (a weaker test, with two analyses) and for SiO_2, FeO and MgO in the olivine; the larger deviations (glass MnO −25 %, F +28 %, P_2O_5 −5 %; olivine Al_2O_3 +6 %, Cr_2O_3 +13 %) are all in imprecise oxides. The correction factors show where the raw data were biased (Fig. 2a, b): olivine CaO read 10.6 % high, calibrated on wollastonite with 47.7 wt% CaO and applied at 0.1–0.4 wt%, a large extrapolation across matrices; glass MgO read 6.1 % high. In the LA-ICP-MS session, 39 of 40 elements of the KL2-G test split lie within ±5 % of their reference (largest: Cs, 7.5 %), and BCR-2G, the standard closest to the inclusions in matrix and concentration, agrees within ±5 % for 31 of 39 elements (median 1.8 %; Sr, Ba, Zr, Nb and Pb within 1.6 %, REE within 8.2 %). The ultramafic GOR132-G, GOR128-G and BM90/21-G, whose incompatible elements lie close to the detection limit, deviate more (median 5.3, 5.5 and 10.1 %; Table A4). With run 70 included, the median BCR-2G deviation would be 12 %

Table 2: Quantitative: 2RSD $\leq 20\,\%$ and accuracy within $\pm 15\,\%$ (the LA-ICP-MS criteria, applied to all sessions), with sample concentrations well above the detection limit (DL).

	Quantitative	Caveat only (reason)
EPMA glass	SiO_2, TiO_2, Al_2O_3, FeO, MgO, CaO, Na_2O, K_2O, Cl	MnO (precision, accuracy); P_2O_5 (precision); SO_3 (samples near DL, 18 of 81 below); F (VG-2 below DL, not quantitative)
EPMA olivine	MgO, FeO, Fo, MnO, NiO, CaO, Al_2O_3	SiO_2 (drift, $\pm 3\,\%$); Cr_2O_3 (precision; near DL, 36 of 113 below)
LA-ICP-MS	Li, Na, Al, K, Ti, V, Rb, Sr, Y, Zr, Nb, Ba, REE, Hf, Ta, Pb, Th, U	B, P, As, Cs (step 6); Mg, Si, Mn, Fe, Sc of inclusions (host olivine); Pb of 9 analyses (contamination)

and its RSTD 58 %. GLITTER also used 50 900 µg/g Ca for BCR-2G against 50 457.5 in `Standards.xlsx` (+0.9 %, well inside the precision).

Drift. SC-Olivine SiO_2 rises by 1.5 wt% during the session, and a single factor cannot follow it: the CF (0.982) raises all SiO_2 by 1.8 %, too little for the early analyses and too much for the late ones (corrected SC-Olivine runs from about $-1\,\%$ to $+3\,\%$ of its reference). As a result, 42 of 113 olivine totals exceed 102 wt% after correction (11 before), 32 of them in the second half of the session and 40 in sample C21MI (Fig. 1d). This is reported rather than normalised away: Fo and all ratios of MgO and FeO are unaffected, and olivine SiO_2 is uncertain by up to 3 %. In the LA-ICP-MS session the fitted CF changes over the session by more than 0.10 only for B (+0.29), As (+0.21), U (+0.12) and Cs (−0.12), and by about 0.05 or less for every other element: the KL2-G bracketing in GLITTER had already removed most of the drift.

Detection limits and reportability (Table 2). VG-2 contains 0.032 wt% F, below the median detection limit of the session (0.037 wt%), so the F correction factor (1.49) is meaningless and F is not quantitative. SO_3 is below the detection limit in 18 of 81 glass spots and Cr_2O_3 in 36 of 113 olivine spots; SC-Olivine itself has only 0.005–0.022 wt% Cr_2O_3, near the detection limit (0.007 wt%), hence its 70 % precision. No inclusion value is below the MDL. B (RSTD 27 %, BCR-2G −30 %), P (−19.5 %), As (RSTD 48 %) and Cs (RSTD 27 %) are flagged and reported with a caveat only.

Problematic inclusion analyses. Pb varies by almost three orders of magnitude (2.6 to 1860 µg/g), while its neighbours in the multi-element pattern, Ce and Pr, vary by factors of 7 and 3.5. The Pb values form two groups with nothing in between, 38 analyses with 2.6–22 µg/g and nine with 56–1860 µg/g; these nine analyses (eight inclusions) are flagged as probable Pb contamination, and their Pb is reported but not interpreted. The major-element oxide sums of the inclusions range from 67 to 416 wt% (13 of 47 within 95–105 wt%) and follow Si, Mg, Fe and Mn (r = 0.96–0.99) and partly Sc (0.69) and Li (0.39), which olivine also carries, but no other element ($|r| \leq 0.27$): in many inclusions the laser pit reached the host olivine. Olivine contains almost no Ca and no incompatible elements, so their ratio to the internal standard, and hence their concentrations, are unaffected, whereas Mg, Si, Mn, Fe and Sc of the inclusions are not interpreted.

4 Methodology (Manuscript)

Major and minor elements in basaltic glasses and olivine were analysed by wavelength-dispersive electron microprobe (JEOL) at 15 kV, with CITZAF matrix corrections (**?**). Glasses were measured with a 5 µm beam at 10 nA, counting 10–30 s on peak and 10–15 s on background; olivine with a focused beam at 100 nA, 20–60 s on peak and 10–30 s on background. The instrument was calibrated on VGA99, scapolite, KE3 and rhodonite (glass) and on San Carlos olivine, wollastonite, rhodonite, corundum, Cr_2O_3 and NiO (olivine). Session offsets were removed with one factor per oxide derived from a standard analysed throughout the session, VG-2 for the glasses (n = 6) and San Carlos olivine for the olivines (n = 15). Accuracy, estimated from standard analyses withheld from the correction, is better than $\pm 1.2\,\%$ relative for SiO_2, TiO_2, Al_2O_3, FeO, MgO and CaO in the glass and SiO_2, FeO and MgO in olivine, with reproducibilities of 0.6–3.1 % (2RSD); Na_2O, K_2O, Cl and olivine MnO, NiO and CaO reproduce at 4–12 %, glass MnO and P_2O_5 at 26–38 %. Olivine SiO_2 is uncertain by up to 3 % because of drift during the session. F, SO_3 and Cr_2O_3 are close to their detection limits and are not reported quantitatively.

Trace elements in melt inclusions were determined by LA-ICP-MS and reduced with GLITTER (Griffin et al., 2008), using KL2-G as calibration standard and ^{43}Ca as internal standard. Residual drift was removed with a linear fit of the KL2-G response against the analysis sequence (11 analyses). Five further KL2-G analyses withheld from the fit agree with GeoReM values (Jochum et al., 2005) within $\pm 5\,\%$ for 39 of 40 elements, and BCR-2G (n = 3), measured as an unknown together with GOR128-G, GOR132-G and BM90/21-G, within $\pm 5\,\%$ for 31 of 39 elements, including Sr, Ba, Zr, Nb and Pb (within 1.6 %) and the REE (within 8.2 %). KL2-G reproducibility is 4–17 % (2RSD) for the reported elements. B, P, As and Cs are not reported quantitatively, nor are Mg, Si, Mn, Fe and Sc of the inclusions, which include host olivine. Data are normalised to the CI chondrite and primitive mantle values of Sun and McDonough (1989).

5 Short presentation of the data

Glasses. The glasses are alkaline basalts, with 44.2–47.5 wt% SiO_2, 4.3–7.6 wt% $Na_2O + K_2O$ and 3.1–3.9 wt% TiO_2 at 3.6–5.3 wt% MgO (sample means; Fig. 3). CaO falls with MgO (r = 0.95 between sample means) while Al_2O_3 ($r = -0.61$) and the alkalis ($r = -0.94$) rise, the pattern expected when clinopyroxene, with or without olivine, crystallises and plagioclase is not yet a major phase; TiO_2 and FeO show no simple trend. BatokeCS1-3 is the most evolved and most heterogeneous sample, and in 1982-CS2-L3A three spots have 3.9–4.6 wt% Na_2O against 2.3–2.8 wt% in the other six.

Olivines. Forsterite ranges from 78.6 to 86.8 mol%: 83.5–86.8 in the two C20-S2 samples and 78.6–85.6 in C21MI (Fig. 4). As Fo decreases, NiO falls (r = 0.91 with Fo) and MnO rises ($r = -0.96$), as expected for olivine fractionation, Ni being compatible and Mn incompatible in olivine; CaO shows no clear trend (r = 0.31). Eleven grains vary by more than 0.5 mol% Fo (2SD), i.e. are zoned. Olivine in Fe–Mg equilibrium with the C21 glass would be Fo 64–72, well below the C21MI grains (78.6–85.6): if glass and olivine come from the same sample, the olivine grew from a more

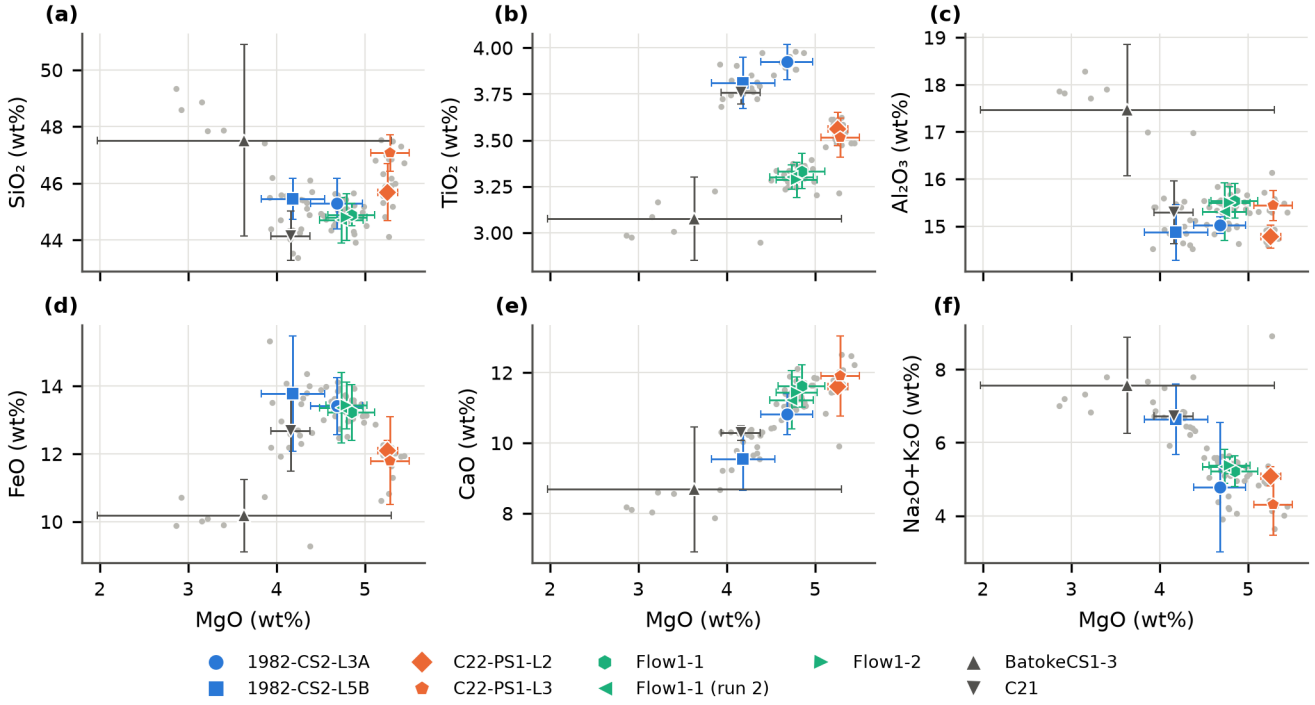


Figure 3: Harker diagrams of the basaltic glasses: spots (grey) and sample means ± 2SD (open symbols). Analyses No. 49 and 65 are excluded.

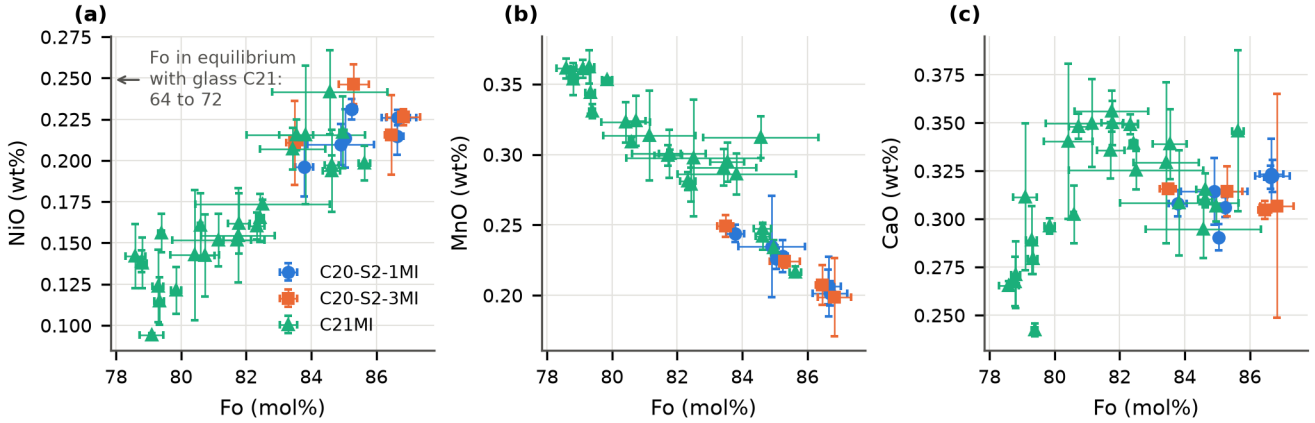


Figure 4: NiO, MnO and CaO against forsterite in the olivine grains (means ± 2SD). The arrow in (a) gives the Fo of olivine in Fe–Mg equilibrium with the mean C21 glass ($K_D = 0.30 \pm 0.03$, Roeder and Emslie, 1970; 85–100 % of the iron as Fe^{2+}).

primitive melt than the one quenched to glass.

Melt inclusions. All inclusions have steep, light-REE-enriched patterns (Fig. 5a), with $(La/Yb)_N$ of 4.8–39.7 (median 17.8) and no europium anomaly (Eu/Eu^* 0.88–1.34, median 1.00). Normalised to primitive mantle (Fig. 5b) they show a positive Nb–Ta anomaly (Nb_N/La_N median 1.32), negative K and Pb anomalies (K/K* 0.41) and low Zr/Nb (median 3.7): the signature of ocean-island or within-plate alkaline basalts, consistent with the alkaline glasses. The four suites share the same pattern shape; MC59 and MC99U are more enriched in the light REE ($(La/Yb)_N$ 22 and 30, against 17–18 in MC22 and MC82).

6 Conclusion

The same procedure gave different outcomes in the three sessions. A single correction factor per element works when the offset of the standard is constant through the session: in the glass, every held-out oxide reproduces its reference within its own precision. It fails for an element that drifts: olivine SiO_2 rose by 1.5 wt% during the session, and one factor for the whole session turned 11 olivine totals above 102 wt% into 42. A time-dependent factor, as fitted to KL2-G in the LA-ICP-MS session, is the remedy, and the KL2-G analyses withheld from the fit show that it keeps the accuracy.

Three further lessons. First, what calibrates what decides the accuracy: the largest bias among the quantitative oxides, olivine CaO (10.6 %), comes from a calibration on wollastonite, far from olivine in matrix and concentration. Second, a single analysis with a wrong internal-standard value turned BCR-2G, the independent standard closest to the inclusions, from a 1.8 % accuracy check into an apparently useless one (12 %); it was found only by checking the Ca value used for each analysis. Third, the detection limit and the ablated volume decide what can be reported: F, SO_3 and Cr_2O_3 in the EPMA data, B, P, As and Cs in the LA-ICP-MS data, and the host-dominated major elements of the inclusions are given with caveats only.

The data describe alkaline basaltic melts evolving by clinopyroxene (and olivine) fractionation, olivines that grew from a more primitive melt than the glass, and melt inclusions

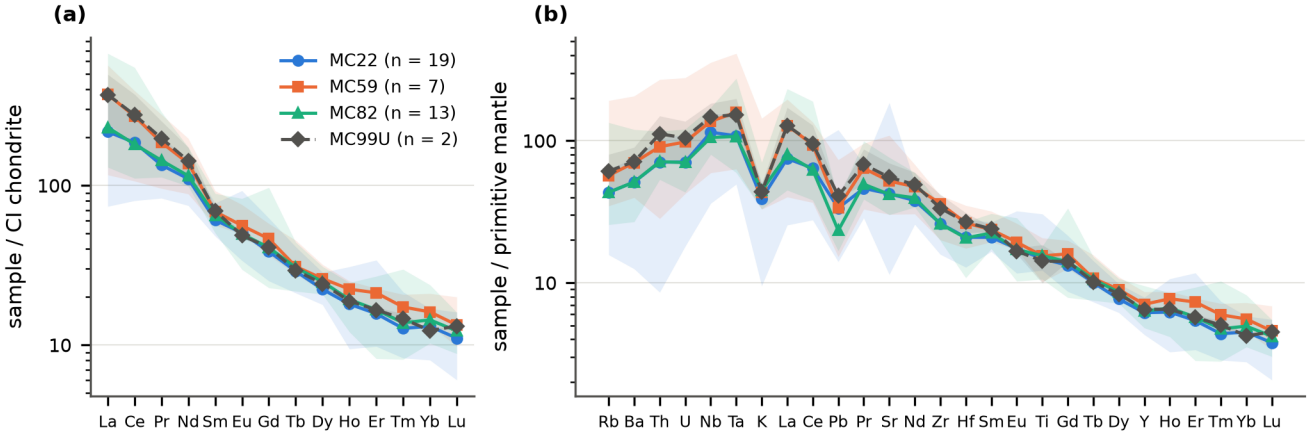


Figure 5: Melt-inclusion trace elements: median (line) and range (shaded) of each suite, normalised to (a) CI chondrite and (b) primitive mantle (Sun and McDonough, 1989). Flagged elements and the Pb of the eight inclusions with probable Pb contamination are omitted; MC99U (two inclusions) in grey.

with the light-REE-enriched, Nb–Ta-rich signature of within-plate basalts. The main limitation lies outside the processing: the EPMA CaO used as internal standard for the inclusions is not part of this dataset, and no sample was measured by both techniques, so the two could not be cross-checked.

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Appendix: full data tables

Sample means with their standards (Tables A1 to A3), and the accuracy and precision of every LA-ICP-MS standard (Table A4). The same tables are provided as an Excel file (`tables_appendix.xlsx`, one sheet per table).

Table A1: Basaltic glasses (EPMA): sample means ± 2SD in wt%, and the VG-2 standard (part 1 of 2)

	n	SiO_2	TiO_2	Al_2O_3	FeO	MnO	MgO	CaO
1982-CS2-L3A	9	45.29 ± 0.90	3.923 ± 0.096	15.02 ± 0.18	13.42 ± 0.84	0.220 ± 0.073	4.68 ± 0.29	10.83 ± 0.59
1982-CS2-L5B	9	45.45 ± 0.72	3.81 ± 0.14	14.87 ± 0.59	13.8 ± 1.7	0.257 ± 0.073	4.19 ± 0.36	9.55 ± 0.88
BatokeCS1-3	8	47.5 ± 3.4	3.08 ± 0.23	17.5 ± 1.4	10.2 ± 1.1	0.170 ± 0.052	3.6 ± 1.7	8.7 ± 1.8
C21	9	44.15 ± 0.87	3.760 ± 0.064	15.29 ± 0.66	12.7 ± 1.2	0.240 ± 0.025	4.16 ± 0.22	10.29 ± 0.21
C22-PS1-L2	9	45.7 ± 1.0	3.561 ± 0.091	14.78 ± 0.24	12.09 ± 0.32	0.196 ± 0.053	5.26 ± 0.11	11.60 ± 0.20
C22-PS1-L3	8	47.07 ± 0.65	3.51 ± 0.11	15.44 ± 0.32	11.8 ± 1.3	0.194 ± 0.088	5.28 ± 0.22	11.9 ± 1.1
Flow1-1	9	44.90 ± 0.39	3.334 ± 0.094	15.53 ± 0.37	13.22 ± 0.82	0.218 ± 0.067	4.85 ± 0.26	11.62 ± 0.60
Flow1-1 (run 2)	9	44.70 ± 0.80	3.301 ± 0.066	15.31 ± 0.61	13.4 ± 1.0	0.25 ± 0.11	4.73 ± 0.25	11.23 ± 0.82
Flow1-2	9	44.81 ± 0.83	3.286 ± 0.095	15.50 ± 0.33	13.43 ± 0.70	0.245 ± 0.056	4.79 ± 0.23	11.44 ± 0.43
VG-2 measured, uncorrected	6	51.42 ± 0.45	1.858 ± 0.043	14.016 ± 0.090	11.88 ± 0.30	0.213 ± 0.081	7.12 ± 0.15	11.30 ± 0.12
VG-2 reference		50.81	1.85	14.06	11.84	0.22	6.71	11.12
Correction factor (measured / reference)		1.012	1.005	0.997	1.003	0.970	1.061	1.016
Precision, 2RSD (%)		0.9	2.3	0.6	2.5	37.8	2.1	1.1
Accuracy, hold-out test (%)		−0.7	+0.9	−0.5	+0.1	−25.0	−1.2	−0.5

Table A1: Basaltic glasses (EPMA): sample means ± 2SD in wt%, and the VG-2 standard (part 2 of 2)

	n	Na_2O	K_2O	P_2O_5	SO_3	Cl	F	Total
1982-CS2-L3A	9	3.1 ± 1.8	1.68 ± 0.15	0.71 ± 0.12	0.037 ± 0.039	0.122 ± 0.013	0.090 ± 0.035	99.12 ± 0.60
1982-CS2-L5B	9	4.63 ± 0.78	2.00 ± 0.26	0.785 ± 0.055	0.036 ± 0.019	0.132 ± 0.030	0.129 ± 0.054	99.61 ± 0.94
BatokeCS1-3	8	5.3 ± 1.3	2.28 ± 0.47	1.32 ± 0.15	0.017 ± 0.023	0.07 ± 0.14	0.167 ± 0.057	99.95 ± 0.82
C21	9	4.61 ± 0.11	2.121 ± 0.077	1.050 ± 0.080	0.113 ± 0.038	0.127 ± 0.046	0.113 ± 0.081	98.9 ± 2.0
C22-PS1-L2	9	3.86 ± 0.20	1.220 ± 0.079	0.568 ± 0.048	0.043 ± 0.023	0.079 ± 0.018	0.086 ± 0.038	99.0 ± 1.1
C22-PS1-L3	8	3.38 ± 0.61	0.93 ± 0.60	0.546 ± 0.091	0.047 ± 0.029	0.075 ± 0.016	0.058 ± 0.040	100.25 ± 0.71
Flow1-1	9	3.49 ± 0.35	1.72 ± 0.15	0.76 ± 0.11	0.036 ± 0.017	0.154 ± 0.023	0.046 ± 0.046	99.9 ± 1.2
Flow1-1 (run 2)	9	3.55 ± 0.34	1.79 ± 0.21	0.77 ± 0.11	0.045 ± 0.027	0.153 ± 0.033	0.038 ± 0.051	99.2 ± 1.6
Flow1-2	9	3.63 ± 0.24	1.75 ± 0.12	0.771 ± 0.079	0.040 ± 0.015	0.147 ± 0.079	0.059 ± 0.077	99.9 ± 1.1
VG-2 measured, uncorrected	6	2.75 ± 0.19	0.195 ± 0.013	0.189 ± 0.050	0.360 ± 0.019	0.0327 ± 0.0039	0.048 ± 0.072	
VG-2 reference		2.62	0.19	0.2	0.3571	0.03291	0.032	
Correction factor (measured / reference)		1.050	1.028	0.944	1.009	0.994	1.491	
Precision, 2RSD (%)		6.8	6.5	26.5	5.2	12.0	150.2	
Accuracy, hold-out test (%)		+1.4	+3.9	−5.1	−2.9	+0.4	+27.9	

Mean ± 2SD in wt%; n = number of spots. All oxides corrected with VG-2: corrected = raw / CF, CF = VG-2 measured / reference.
Excluded from the means: analyses No. 49, 65; SO_3 of No. 32; Cl of No. 70.
Correction factor outside ±5 % (step 5): MgO (1.061), P_2O_5 (0.944), F (1.491).
Below the detection limit (flagged, kept in the means): SO_3 in 18, F in 12, K_2O in 1, P_2O_5 in 1, Cl in 1 of 81 analyses.
F is not quantitative: the VG-2 reference value (0.032 wt%) is below the median detection limit (0.037 wt%), so its correction factor is unreliable.
Flow1-1 (run 2): the same sample measured twice under one name; both runs agree within 2SD.
VG-2: n = 6, except SiO_2 (n = 5), Al_2O_3 (n = 5), Cl (n = 5) after the outliers removed in step 3. The corrected standard equals its reference by construction.

Table A2: Olivine grains (EPMA): grain means ± 2SD in wt% (Fo in mol%), and the SC-Olivine standard (part 1 of 2)

	n	SiO_2	Al_2O_3	Cr_2O_3	FeO	MnO
C20-S2-1MI_1	3	39.7 ± 1.3	0.031 ± 0.012	0.029 ± 0.012	14.49 ± 0.97	0.235 ± 0.036
C20-S2-1MI_2	3	39.9 ± 1.6	0.041 ± 0.014	0.011 ± 0.013	15.37 ± 0.15	0.2439 ± 0.0062
C20-S2-1MI_3	3	40.1 ± 1.3	0.034 ± 0.021	0.039 ± 0.012	12.91 ± 0.21	0.206 ± 0.012
C20-S2-1MI_4	3	40.3 ± 1.5	0.032 ± 0.016	0.0353 ± 0.0080	12.89 ± 0.47	0.2016 ± 0.0086
C20-S2-1MI_5	3	39.9 ± 1.2	0.031 ± 0.010	0.037 ± 0.055	14.39 ± 0.11	0.2259 ± 0.0073
C20-S2-1MI_6	3	39.515 ± 0.082	0.0287 ± 0.0045	0.0365 ± 0.0099	12.90 ± 0.35	0.206 ± 0.021
C20-S2-1MI_7	3	39.8 ± 1.3	0.032 ± 0.012	0.028 ± 0.019	14.19 ± 0.21	0.228 ± 0.011
C20-S2-3MI_1	3	41.16 ± 0.19	0.039 ± 0.012	0.039 ± 0.025	12.65 ± 0.47	0.199 ± 0.028
C20-S2-3MI_2	3	40.87 ± 0.21	0.0338 ± 0.0091	0.0254 ± 0.0097	14.09 ± 0.30	0.2242 ± 0.0047
C20-S2-3MI_3	3	40.723 ± 0.093	0.0347 ± 0.0060	0.015 ± 0.018	15.76 ± 0.29	0.2492 ± 0.0076
C20-S2-3MI_4	3	40.947 ± 0.020	0.0281 ± 0.0040	0.035 ± 0.019	13.08 ± 0.18	0.207 ± 0.014
C21MI_1	3	39.4 ± 1.4	0.0316 ± 0.0041	0.010 ± 0.013	17.36 ± 0.31	0.3009 ± 0.0026
C21MI_10	3	40.65 ± 0.19	0.0262 ± 0.0035	0.016 ± 0.014	15.4 ± 1.7	0.286 ± 0.015
C21MI_11	3	40.99 ± 0.21	0.0227 ± 0.0048	0.016 ± 0.017	15.72 ± 0.51	0.295 ± 0.014
C21MI_12	3	41.03 ± 0.18	0.0226 ± 0.0086	0.012 ± 0.024	15.82 ± 0.87	0.291 ± 0.013
C21MI_13	3	40.75 ± 0.37	0.033 ± 0.021	0.0144 ± 0.0054	16.7 ± 1.9	0.298 ± 0.041
C21MI_14	3	40.1 ± 1.1	0.0300 ± 0.0082	0.0041 ± 0.0072	19.48 ± 0.13	0.344 ± 0.015
C21MI_15	3	39.475 ± 0.059	0.026 ± 0.011	0.0022 ± 0.0057	19.00 ± 0.21	0.3531 ± 0.0017
C21MI_16	3	39.22 ± 0.32	0.0272 ± 0.0066	0.0028 ± 0.0050	18.26 ± 0.30	0.324 ± 0.018
C21MI_17	3	40.62 ± 0.21	0.0282 ± 0.0094	0.012 ± 0.013	17.3 ± 1.0	0.301 ± 0.017
C21MI_18	3	40.74 ± 0.19	0.0306 ± 0.0060	0.0084 ± 0.0040	16.89 ± 0.26	0.2813 ± 0.0064
C21MI_19	3	39.42 ± 0.28	0.0286 ± 0.0077	0.0061 ± 0.0070	19.403 ± 0.095	0.3310 ± 0.0050
C21MI_2	3	39.4 ± 1.1	0.0252 ± 0.0045	0.009 ± 0.015	17.9 ± 1.2	0.313 ± 0.032
C21MI_20	3	40.516 ± 0.062	0.0298 ± 0.0057	0.0319 ± 0.0076	14.44 ± 0.12	0.2337 ± 0.0054
C21MI_21	3	41.2 ± 1.1	0.028 ± 0.011	0.033 ± 0.011	13.83 ± 0.18	0.2166 ± 0.0040
C21MI_22	3	40.3 ± 1.5	0.037 ± 0.017	0.0212 ± 0.0091	14.78 ± 0.27	0.2472 ± 0.0033
C21MI_23	3	40.42 ± 0.17	0.0287 ± 0.0077	0.006 ± 0.011	18.399 ± 0.031	0.3090 ± 0.0035
C21MI_24	4	39.8 ± 1.2	0.032 ± 0.012	0.0006 ± 0.0014	19.70 ± 0.39	0.3610 ± 0.0064
C21MI_25	3	40.0 ± 1.3	0.0311 ± 0.0028	0.018 ± 0.017	16.79 ± 0.11	0.279 ± 0.012
C21MI_26	3	40.34 ± 0.42	0.0292 ± 0.0046	0.0079 ± 0.0033	17.35 ± 0.40	0.2995 ± 0.0073
C21MI_3	3	39.8 ± 1.3	0.0219 ± 0.0018	0.021 ± 0.016	14.8 ± 1.6	0.312 ± 0.015
C21MI_4	4	39.9 ± 1.4	0.0284 ± 0.0087	0.0037 ± 0.0056	19.93 ± 0.11	0.354 ± 0.011
C21MI_5	3	39.6 ± 1.4	0.0313 ± 0.0064	0.0052 ± 0.0092	20.15 ± 0.40	0.3612 ± 0.0071
C21MI_6	3	38.70 ± 0.19	0.030 ± 0.016	0.006 ± 0.011	19.972 ± 0.040	0.3579 ± 0.0044
C21MI_7	3	40.3 ± 1.4	0.0288 ± 0.0035	0.0155 ± 0.0027	14.80 ± 0.19	0.2419 ± 0.0095
C21MI_8	3	39.7 ± 1.4	0.0294 ± 0.0039	0.005 ± 0.012	18.50 ± 0.64	0.323 ± 0.014
C21MI_9	3	39.9 ± 1.1	0.0261 ± 0.0064	0.0053 ± 0.0049	19.50 ± 0.16	0.362 ± 0.012
SC-Olivine measured, uncorrected	15	40.1 ± 1.3	0.0333 ± 0.0058	0.0141 ± 0.0099	9.51 ± 0.14	0.1433 ± 0.0085
SC-Olivine reference		40.82	0.033	0.014	9.546	0.14
Correction factor (measured / reference)		0.982	1.008	1.011	0.996	1.024
Precision, 2RSD (%)		3.1	17.4	69.7	1.5	5.9
Accuracy, hold-out test (%)		−0.1	+5.9	+12.5	−0.7	−3.3

Table A2: Olivine grains (EPMA): grain means ± 2SD in wt% (Fo in mol%), and the SC-Olivine standard (part 2 of 2)

	n	MgO	CaO	NiO	Total	Fo
C20-S2-1MI_1	3	45.73 ± 0.57	0.314 ± 0.017	0.210 ± 0.013	100.72 ± 0.84	84.9 ± 1.0
C20-S2-1MI_2	3	44.51 ± 0.52	0.3082 ± 0.0069	0.196 ± 0.018	100.6 ± 1.8	83.78 ± 0.27
C20-S2-1MI_3	3	46.95 ± 0.17	0.3216 ± 0.0060	0.215 ± 0.011	100.8 ± 1.1	86.64 ± 0.19
C20-S2-1MI_4	3	47.11 ± 0.48	0.3223 ± 0.0083	0.2259 ± 0.0013	101.2 ± 1.6	86.69 ± 0.53
C20-S2-1MI_5	3	45.85 ± 0.18	0.2905 ± 0.0067	0.213 ± 0.018	100.9 ± 1.0	85.032 ± 0.057
C20-S2-1MI_6	3	47.01 ± 0.24	0.323 ± 0.018	0.2261 ± 0.0047	100.24 ± 0.21	86.66 ± 0.36
C20-S2-1MI_7	3	46.00 ± 0.18	0.3062 ± 0.0053	0.2311 ± 0.0062	100.86 ± 0.96	85.24 ± 0.15
C20-S2-3MI_1	3	46.77 ± 0.34	0.307 ± 0.058	0.2263 ± 0.0049	101.390 ± 0.071	86.83 ± 0.51
C20-S2-3MI_2	3	45.91 ± 0.79	0.314 ± 0.013	0.246 ± 0.012	101.72 ± 0.37	85.31 ± 0.46
C20-S2-3MI_3	3	44.674 ± 0.053	0.3158 ± 0.0015	0.211 ± 0.026	101.98 ± 0.28	83.48 ± 0.24
C20-S2-3MI_4	3	46.87 ± 0.13	0.3047 ± 0.0045	0.216 ± 0.024	101.692 ± 0.062	86.46 ± 0.19
C21MI_1	3	43.71 ± 0.20	0.350 ± 0.012	0.162 ± 0.018	101.4 ± 1.3	81.78 ± 0.34
C21MI_10	3	44.9 ± 1.2	0.308 ± 0.027	0.216 ± 0.042	101.82 ± 0.34	83.8 ± 1.8
C21MI_11	3	44.77 ± 0.25	0.339 ± 0.018	0.2156 ± 0.0094	102.374 ± 0.050	83.54 ± 0.52
C21MI_12	3	44.70 ± 0.80	0.329 ± 0.042	0.207 ± 0.013	102.41 ± 0.19	83.4 ± 1.0
C21MI_13	3	44.1 ± 1.4	0.3254 ± 0.0100	0.1734 ± 0.0029	102.39 ± 0.34	82.5 ± 2.1
C21MI_14	3	41.95 ± 0.11	0.2793 ± 0.0079	0.115 ± 0.014	102.3 ± 1.1	79.33 ± 0.15
C21MI_15	3	42.236 ± 0.090	0.2964 ± 0.0040	0.121 ± 0.014	101.51 ± 0.35	79.85 ± 0.15
C21MI_16	3	42.91 ± 0.10	0.3481 ± 0.0065	0.142 ± 0.025	101.23 ± 0.15	80.73 ± 0.29
C21MI_17	3	43.46 ± 0.76	0.356 ± 0.011	0.155 ± 0.029	102.229 ± 0.030	81.7 ± 1.1
C21MI_18	3	44.104 ± 0.041	0.3491 ± 0.0052	0.1603 ± 0.0080	102.560 ± 0.071	82.32 ± 0.24
C21MI_19	3	41.921 ± 0.069	0.2424 ± 0.0033	0.156 ± 0.012	101.50 ± 0.39	79.387 ± 0.092
C21MI_2	3	43.2 ± 1.1	0.350 ± 0.023	0.152 ± 0.016	101.4 ± 1.2	81.1 ± 1.4
C21MI_20	3	45.780 ± 0.064	0.3067 ± 0.0080	0.217 ± 0.022	101.55 ± 0.17	84.97 ± 0.10
C21MI_21	3	46.25 ± 0.13	0.346 ± 0.042	0.198 ± 0.010	102.1 ± 1.2	85.64 ± 0.18
C21MI_22	3	45.63 ± 0.14	0.3152 ± 0.0086	0.194 ± 0.025	101.5 ± 1.6	84.62 ± 0.27
C21MI_23	3	42.86 ± 0.26	0.302 ± 0.015	0.161 ± 0.019	102.48 ± 0.38	80.590 ± 0.070
C21MI_24	4	41.802 ± 0.099	0.311 ± 0.038	0.0943 ± 0.0009	102.1 ± 1.0	79.09 ± 0.36
C21MI_25	3	44.18 ± 0.12	0.3392 ± 0.0018	0.165 ± 0.015	101.8 ± 1.2	82.43 ± 0.11
C21MI_26	3	43.53 ± 0.26	0.336 ± 0.015	0.1517 ± 0.0015	102.04 ± 0.24	81.73 ± 0.43
C21MI_3	3	45.5 ± 1.2	0.295 ± 0.015	0.241 ± 0.025	101.0 ± 1.6	84.6 ± 1.8
C21MI_4	4	41.57 ± 0.18	0.271 ± 0.017	0.138 ± 0.015	102.1 ± 1.3	78.80 ± 0.14
C21MI_5	3	41.517 ± 0.057	0.2654 ± 0.0024	0.142 ± 0.019	102.1 ± 1.2	78.60 ± 0.31
C21MI_6	3	41.590 ± 0.083	0.268 ± 0.013	0.1389 ± 0.0066	101.06 ± 0.25	78.778 ± 0.053
C21MI_7	3	45.69 ± 0.16	0.3104 ± 0.0044	0.1974 ± 0.0058	101.6 ± 1.5	84.63 ± 0.21
C21MI_8	3	42.62 ± 0.61	0.340 ± 0.040	0.143 ± 0.040	101.7 ± 1.2	80.42 ± 0.75
C21MI_9	3	41.875 ± 0.070	0.289 ± 0.018	0.124 ± 0.022	102.1 ± 1.2	79.29 ± 0.17
SC-Olivine measured, uncorrected	15	49.74 ± 0.29	0.1018 ± 0.0040	0.359 ± 0.023		
SC-Olivine reference		49.43	0.092	0.37		
Correction factor (measured / reference)		1.006	1.106	0.969		
Precision, 2RSD (%)		0.6	4.0	6.5		
Accuracy, hold-out test (%)		−0.1	+2.1	−0.2		

Mean ± 2SD in wt%, Fo in mol%; n = number of spots per grain. Corrected with SC-Olivine: corrected = raw / CF, CF = SC-Olivine measured / reference.
Correction factor outside ±5 % (step 5): CaO (1.106).
Below the detection limit (flagged, kept in the means): Cr_2O_3 in 36 of 113 analyses.
SC-Olivine: n = 15, except Al_2O_3 (n = 14), Cr_2O_3 (n = 14), NiO (n = 14) after the outliers removed in step 3. The corrected standard equals its reference by construction.

Table A3: Melt inclusions (LA-ICP-MS): concentrations in μg/g, and the standards measured as unknowns (part 1 of 4)

	n	^{7}Li	^{11}B *	^{23}Na	^{24}Mg^{h}	^{27}Al	^{29}Si^{h}	^{31}P *	^{39}K	^{43}Ca (IS)	^{45}Sc^{h}	^{47}Ti
MC22-1-16	1	5.23	1.49	20858	235808	68089	283274	2488	9764	88337	38.2	19584
MC22-1-20	1	3.76	1.62	19745	28983	70799	167672	1695	8132	96485	36.9	25444
MC22-1-21	1	4.58	1.15	15560	108370	59165	188596	1158	5739	94269	33.8	24224
MC22-1-26	2	7.37	2.01	30883	31155	79436	204773	2400	11231	95484	30.5	18952
MC22-1-27	1	5.83	1.87	25460	21996	70570	173475	2225	12717	75615	18	16209
MC22-2-1	1	5.32	1.14	17231	525951	77019	488571	1753	8271	90052	47.3	16545
MC22-2-23	1	4.67	1.99	25000	31391	78858	202442	2961	10534	95556	34.3	20918
MC22-2-24	1	6.96	2.51	22099	172970	56857	253868	2637	10857	83048	24.5	15922
MC22-2-27	1	4.08	2.42	23398	30832	70436	187650	1980	9936	91696	29.2	17797
MC22-2-28	1	3.92	2.05	22445	47701	79354	178341	2492	11837	86836	33.3	19589
MC22-2-29	2	7.26	2.81	22473	374866	94593	374447	1692	7918	104346	40.9	21327
MC22-2-30	1	8.98	3.92	20896	944051	77002	780747	1476	8556	92697	62.2	16144
MC22-2-31	1	6.95	1.08	25379	42372	91702	213787	311	2379	105204	42	39646
MC22-2-32	1	4.03	1.71	22344	56643	77506	200473	1764	8862	84692	27.3	17536
MC22-2-35	1	5.49	3.69	22550	342829	80200	361478	1637	7542	86050	38.3	19201
MC22-2-40	1	5.01	2.44	26829	36269	78800	209437	2314	10019	86479	31.4	18926
MC22-2-5	1	3.69	1.04	23307	31967	71967	169624	1867	9975	88194	31.8	19522
MC22-2-8	1	3.16	6.81	12092	137655	45753	207310	1872	7949	93483	28.2	16562
MC22-2-9	1	4.89	1.39	24999	35204	75363	196845	2526	12643	85907	30	20236
MC59-1-2	1	9.94	2.79	45008	59689	106669	251226	3782	22690	64752	16.1	20116
MC59-1-3	2	4.36	2.68	29740	131039	101791	255590	2377	10841	109814	32.1	19466
MC59-1-7	1	5.75	2.31	33885	27623	87940	218556	2096	9895	103274	29.8	26872
MC59-2-1	1	4.36	2.46	27558	21336	82101	188739	4372	8563	114138	34.8	23363
MC59-2-6	1	6.97	3.97	33357	40386	97090	223107	6147	35912	76830	22.4	13191
MC59-2-8	1	4	0.75	24006	117824	89734	236069	1645	8221	113280	36.6	19025
MC59-9-1	3	5.97	3.34	34259	57492	89701	258139	2162	11374	111041	38.3	21087
MC82-11	1	4.49	2.47	24591	68531	90928	227028	2800	12261	112065	38.4	23216
MC82-12	2	6.82	2.33	28973	28373	74048	196292	2561	11752	89552	27.6	19866
MC82-19	1	5.59	2.59	21433	296934	69948	339857	2369	9781	97628	40.2	18911
MC82-21	1	4.23	2.1	20326	22573	73113	193684	2085	8267	97700	29.3	19793
MC82-28	1	6.94	11.3	62876	47936	147896	285962	3766	17599	71970	15	20191
MC82-29	1	4.53	2.24	25369	24962	82973	203581	2142	9877	97914	35	21524
MC82-32	1	2.43	1.18	14691	16341	50556	142372	1944	11635	90910	23.7	13779
MC82-35	1	7.23	3.33	34481	26096	79507	207851	3160	16285	71613	17.5	23472
MC82-37	1	4.73	1.45	25165	36979	79810	181883	1877	8962	91124	29.5	22789
MC82-39	1	5.22	1.72	26633	35421	79452	196438	1974	9905	86407	29.3	21126
MC82-40	1	4.88	2.39	27263	33471	88176	206290	2360	11120	96628	34.4	20653
MC82-5	1	7.56	11.6	32543	24012	84720	186761	3430	14237	63751	16.1	18459
MC82-6	1	4.08	1.71	22002	22803	77896	168980	1859	8842	91982	27.9	19251
MC99U-10	1	5.06	1.53	20980	130897	88084	257184	1993	9518	115639	41.5	24048
MC99U-7	1	6.91	4.62	26172	178679	87594	299260	3946	12281	97199	38.2	12950
BCR-2G measured	3	7.77 ± 0.30	4.2 ± 1.3	23320 ± 1156	21072 ± 895	71676 ± 1438	242459 ± 10822	1300 ± 109	14419 ± 565	50900	34.6 ± 1.1	13746 ± 40
BCR-2G reference		9.0 ± 1.0	6.0 ± 1.0	23962 ± 519	21468 ± 543	70921 ± 2117	254290 ± 1870	1615 ± 44	14445 ± 332	50458 ± 786	33.0 ± 2.0	13605 ± 240
BCR-2G accuracy (%)		−13.7	−30.5	−2.7	−1.8	+1.1	−4.7	−19.5	−0.2		+5.0	+1.0
KL2 test split measured	5	5.15 ± 0.57	2.86 ± 0.88	17780 ± 2148	45147 ± 4394	71306 ± 6377	232280 ± 12203	987 ± 109	3968 ± 308	77902	31.5 ± 1.9	15325 ± 972
KL2 test split reference		5.10 ± 0.50	2.73 ± 0.28	17434 ± 593	44263 ± 543	70392 ± 1059	235125 ± 1402	1013 ± 113	3985 ± 83	77902 ± 1429	31.80 ± 0.90	15343 ± 539
KL2 test split accuracy (%)		+0.9	+4.7	+2.0	+2.0	+1.3	−1.2	−2.5	−0.4		−0.9	−0.1
KL2 precision, RSTD (%)		11.3	26.6	10.4	8.6	7.2	5.5	11.7	5.1		4.5	6.3

Table A3: Melt inclusions (LA-ICP-MS): concentrations in μg/g, and the standards measured as unknowns (part 2 of 4)

	n	^{51}V	^{55}Mn^{h}	^{57}Fe^{h}	^{75}As *	^{85}Rb	^{88}Sr	^{89}Y	^{90}Zr	^{93}Nb	^{133}Cs *	^{138}Ba
MC22-1-16	1	317	2529	150083	0.798	29.8	898	20.5	293	96.2	0.33	409
MC22-1-20	1	384	1137	65711	0.403	22.1	757	30	309	65.2	0.239	315
MC22-1-21	1	296	1860	111638	0.413	11.8	3899	24.6	277	58.9	0.132	376
MC22-1-26	2	289	1535	82477	0.784	37.6	1190	34.7	398	97.3	0.439	485
MC22-1-27	1	218	1295	67440	0.797	40	1255	31.5	374	87.8	0.464	462
MC22-2-1	1	220	3655	222994	0.342	15.9	769	29.6	300	49.7	0.169	191
MC22-2-23	1	358	1194	75972	0.776	34	893	27.2	305	84.3	0.386	400
MC22-2-24	1	252	2841	146533	1.04	33.8	1146	33	361	110	0.405	528
MC22-2-27	1	306	1044	67187	0.584	29.1	819	27	263	81.7	0.348	358
MC22-2-28	1	305	1363	77037	0.826	35.1	1073	25.9	373	105	0.416	493
MC22-2-29	2	341	3218	217429	0.559	24.9	941	35.9	329	61.6	0.278	340
MC22-2-30	1	196	7294	440100	0.951	22.6	730	28	277	55.4	0.205	285
MC22-2-31	1	454	1664	86411	0.357	10	241	35.1	178	25.8	0.12	87.6
MC22-2-32	1	268	1349	84464	0.557	22.9	806	29	273	72.6	0.282	312
MC22-2-35	1	297	3578	223735	0.774	22	773	28.6	267	60.1	0.229	330
MC22-2-40	1	271	1244	78315	0.785	28.2	807	28	292	68.4	0.299	357
MC22-2-5	1	294	1068	65983	0.506	27.5	893	24.9	290	82.6	0.294	388
MC22-2-8	1	247	1704	109513	0.757	16.2	751	19.7	247	88	0.162	257
MC22-2-9	1	315	1309	79702	0.735	33.4	965	27.4	316	96	0.361	423
MC59-1-2	1	178	1772	78923	1.04	69.3	1207	39.3	461	145	0.584	706
MC59-1-3	2	376	1609	98588	0.603	36.1	1189	33.5	345	97.5	0.315	490
MC59-1-7	1	328	1428	84461	0.561	24.4	1020	36.4	404	67.7	0.294	317
MC59-2-1	1	355	981	59290	0.41	22.1	639	28.5	225	42.4	0.284	278
MC59-2-6	1	317	1475	75833	1.48	122	2298	32.1	460	253	1.51	1441
MC59-2-8	1	309	1307	80214	0.707	22.1	851	29.4	282	60.1	0.217	333
MC59-9-1	3	342	1361	94710	1.08	42.7	1096	29	406	114	0.581	513
MC82-11	1	382	1574	94079	0.707	32.8	1168	37.8	394	113	0.451	465
MC82-12	2	287	1348	79649	0.67	27.9	893	27.7	292	79.3	0.317	330
MC82-19	1	316	3655	220191	0.547	25	836	27.7	281	83.5	0.367	334
MC82-21	1	278	1154	64950	0.479	20.2	793	23.4	222	61.8	0.192	275
MC82-28	1	219	1964	122096	2.7	85.1	1609	43.3	388	98.5	0.453	837
MC82-29	1	355	1315	81888	0.636	27.5	926	28.7	292	73.2	0.315	358
MC82-32	1	176	715	41079	0.352	16.3	636	21.9	248	57.1	0.215	188
MC82-35	1	286	1841	97333	0.961	46.6	893	36.3	390	120	0.576	542
MC82-37	1	326	1427	85323	0.514	24.9	885	29	291	70.7	0.279	341
MC82-39	1	333	1442	87228	0.581	27.1	850	27.4	281	71.7	0.28	360
MC82-40	1	348	1364	81376	0.598	30.9	869	26.9	262	75	0.359	384
MC82-5	1	372	1281	43843	1.21	40.9	1536	30.8	357	122	0.465	626
MC82-6	1	286	1112	65853	0.44	25.2	862	29	281	70.5	0.288	303
MC99U-10	1	343	1801	105936	0.566	26.7	866	31.4	282	79.5	0.329	369
MC99U-7	1	224	2145	125196	1.19	50.9	1466	27.7	470	131	0.657	627
BCR-2G measured	3	418 ± 15	1514 ± 96	90811 ± 6563	0.568 ± 0.055	46.80 ± 0.97	338.2 ± 5.3	35.3 ± 1.2	183.4 ± 2.3	12.70 ± 0.64	1.11 ± 0.11	680 ± 13
BCR-2G reference		425 ± 18	1471 ± 77	96386 ± 2332		47.00 ± 0.50	342.0 ± 4.0	35.0 ± 3.0	184 ± 15	12.5 ± 1.0	1.160 ± 0.070	683.0 ± 7.0
BCR-2G accuracy (%)		−1.6	+2.9	−5.8		−0.4	−1.1	+0.8	−0.3	+1.6	−4.4	−0.5
KL2 test split measured	5	315 ± 29	1309 ± 144	85816 ± 6853	0.178 ± 0.084	8.80 ± 0.84	365 ± 20	25.5 ± 1.8	152.3 ± 5.8	15.1 ± 1.0	0.124 ± 0.040	125 ± 12
KL2 test split reference		309 ± 38	1278 ± 70	83172 ± 777	0.17	8.70 ± 0.40	356.0 ± 8.0	25.4 ± 1.1	152.0 ± 5.0	15.00 ± 0.50	0.1150 ± 0.0090	123.0 ± 5.0
KL2 test split accuracy (%)		+1.8	+2.4	+3.2	+4.5	+1.2	+2.5	+0.5	+0.2	+0.9	+7.5	+1.5
KL2 precision, RSTD (%)		9.0	10.4	9.8	48.0	7.7	5.7	5.0	4.1	6.7	26.9	11.6

Table A3: Melt inclusions (LA-ICP-MS): concentrations in μg/g, and the standards measured as unknowns (part 3 of 4)

	n	^{139}La	^{140}Ce	^{141}Pr	^{146}Nd	^{147}Sm	^{151}Eu	^{157}Gd	^{159}Tb	^{163}Dy	^{165}Ho	^{166}Er
MC22-1-16	1	62.2	130	14.7	58	9.27	2.97	6.98	0.878	4.55	0.867	2.48
MC22-1-20	1	42.6	90.3	11.2	52.8	10.3	3.18	7.6	1.06	6.46	1.14	2.99
MC22-1-21	1	37.5	80.1	10.6	50.7	9.84	2.52	6.26	0.84	5.97	1.03	2.47
MC22-1-26	2	74.8	152	17.5	68.3	11.8	3.59	10.4	1.37	7.24	1.26	3.22
MC22-1-27	1	63.5	147	18.8	66.5	9.86	2.96	7.92	1.26	6.13	1.08	2.6
MC22-2-1	1	29.1	81.4	11.9	50.5	7.37	1.96	5.5	1.19	5.12	0.761	1.62
MC22-2-23	1	56.5	114	12.8	49	8.58	2.79	8.02	1.08	5.53	1.04	2.75
MC22-2-24	1	77	164	19	67.4	11.6	3.45	8.56	1.12	5.58	1.19	3.02
MC22-2-27	1	61.3	114	13.3	54.7	8.29	2.78	7.64	0.995	5.66	0.903	2.4
MC22-2-28	1	83.6	158	16.9	67	11.5	3.62	8.81	1.07	5.67	0.953	2.84
MC22-2-29	2	51.4	114	13.1	49.8	9.25	2.93	10.6	1.38	6.42	1.15	3.42
MC22-2-30	1	38.8	87.5	9.98	43.5	8.33	2.37	6.68	1.13	5.27	0.962	2.6
MC22-2-31	1	17.5	49.1	7.91	34.9	11.8	5.37	12.8	1.54	6.85	1.75	5.63
MC22-2-32	1	49.5	99.7	11.9	49.6	8.75	2.65	7.54	1.17	5.95	0.942	2.43
MC22-2-35	1	46.2	96.9	11.1	45.2	7.89	2.55	8.07	1.08	5.28	1.02	2.85
MC22-2-40	1	48.7	104	12.4	50.4	9.94	3.07	8.42	1.12	6.05	1.04	2.77
MC22-2-5	1	62.9	122	13.4	51.4	7.96	2.38	7.32	1.04	5.07	0.909	2.43
MC22-2-8	1	42.7	86.9	12.4	52.6	8.08	2.27	5.18	0.792	4.58	0.535	1.94
MC22-2-9	1	68.2	137	15.8	63.2	10.4	3.13	8.74	1.07	5.74	1.01	2.47
MC59-1-2	1	112	240	26.4	92	13.9	4.39	11.8	1.69	8.11	1.43	3.56
MC59-1-3	2	88	172	18.4	67.8	11.4	3.56	10	1.34	7.45	1.36	3.74
MC59-1-7	1	55.4	130	16.3	69.5	14.2	4.01	11.3	1.58	8.08	1.37	3.51
MC59-2-1	1	27.6	65.2	9.1	41.3	9.4	2.97	8.27	1.09	5.87	1.17	2.76
MC59-2-6	1	134	201	19.1	63.4	10.5	3.25	8.14	1.04	6.21	1.27	3.61
MC59-2-8	1	44.4	99.3	12.5	49.5	8.61	2.88	7.51	1.12	5.53	1.02	2.55
MC59-9-1	3	88.8	166	17.7	64.3	10.1	3.24	9.52	1.16	6.62	1.25	3.28
MC82-11	1	84.6	170	20	80.9	12.5	3.21	9.95	1.49	7.54	1.21	3.3
MC82-12	2	50.9	104	13.6	53.7	9.96	2.75	7.53	1.08	6.32	1.07	2.47
MC82-19	1	57.4	116	13.7	58.3	9.23	2.43	6.64	1.14	6.32	1.11	2.36
MC82-21	1	40.3	90.7	12	46.2	9.07	2.71	6.37	0.975	5.51	0.785	2.17
MC82-28	1	159	336	27.4	76.8	10.6	4.8	19.9	1.65	6.99	1.18	4.46
MC82-29	1	57.6	119	14	54.9	10.4	3.07	8.99	1.17	6.51	1.09	2.73
MC82-32	1	30.8	67.9	10.7	49.6	7.72	1.72	4.69	0.813	5.29	0.684	1.36
MC82-35	1	87.5	170	19.6	75.4	13.6	4.08	11.3	1.4	7.87	1.39	3.73
MC82-37	1	54.8	111	13.1	51.2	10.4	3.08	8.63	1.14	6.86	1.07	2.98
MC82-39	1	54	107	12.9	51.2	9.38	2.85	8.19	1.08	5.77	1.03	2.76
MC82-40	1	53.8	110	13	47.6	9.36	2.88	8.29	1.06	6.12	1.12	2.84
MC82-5	1	98.1	187	21	74.7	11.9	3.5	9.69	1.2	6.46	1.21	2.91
MC82-6	1	52.3	109	13	52	8.74	2.72	8.39	1.2	6.16	0.96	2.5
MC99U-10	1	58.1	111	13.5	52.2	10.2	2.68	8.4	1.12	6.67	1.1	2.84
MC99U-7	1	117	228	24	80.9	11.1	2.95	8.42	1.08	5.63	1.04	2.65
BCR-2G measured	3	24.12 ± 0.58	51.6 ± 1.2	6.42 ± 0.11	28.5 ± 1.3	6.05 ± 0.47	1.988 ± 0.030	6.78 ± 0.49	1.017 ± 0.031	6.28 ± 0.33	1.293 ± 0.083	3.75 ± 0.14
BCR-2G reference		24.70 ± 0.30	53.30 ± 0.50	6.70 ± 0.40	28.90 ± 0.30	6.590 ± 0.070	1.970 ± 0.020	6.710 ± 0.070	1.020 ± 0.080	6.440 ± 0.060	1.270 ± 0.080	3.700 ± 0.040
BCR-2G accuracy (%)		−2.3	−3.2	−4.2	−1.5	−8.2	+0.9	+1.0	−0.3	−2.5	+1.8	+1.4
KL2 test split measured	5	13.3 ± 1.4	32.7 ± 2.4	4.63 ± 0.26	21.4 ± 2.3	5.63 ± 0.67	1.94 ± 0.28	6.16 ± 0.83	0.920 ± 0.092	5.17 ± 0.81	0.94 ± 0.15	2.62 ± 0.42
KL2 test split reference		13.10 ± 0.20	32.40 ± 0.70	4.60 ± 0.10	21.60 ± 0.40	5.540 ± 0.090	1.920 ± 0.040	5.92 ± 0.20	0.890 ± 0.031	5.22 ± 0.12	0.961 ± 0.022	2.540 ± 0.070
KL2 test split accuracy (%)		+1.3	+0.8	+0.7	−1.1	+1.6	+1.0	+4.0	+3.4	−1.0	−2.0	+3.3
KL2 precision, RSTD (%)		11.7	7.3	4.8	7.8	8.5	11.7	10.5	8.6	9.3	12.6	14.0

Table A3: Melt inclusions (LA-ICP-MS): concentrations in μg/g, and the standards measured as unknowns (part 4 of 4)

	n	^{169}Tm	^{172}Yb	^{175}Lu	^{178}Hf	^{181}Ta	^{208}Pb	^{232}Th	^{238}U
MC22-1-16	1	0.253	1.37	0.28	6.76	5.04	12	7.25	2.05
MC22-1-20	1	0.324	2.04	0.365	7.52	3.59	6.14	4.54	1.38
MC22-1-21	1	0.298	1.92	0.381	5.87	2.84	9.09	4.61	1.27
MC22-1-26	2	0.432	2.7	0.365	8.65	5.96	11.8	8.84	2.17
MC22-1-27	1	0.331	2.46	0.279	6.46	4.78	22.1	7.65	1.8
MC22-2-1	1	0.218	2.87	0.287	5.55	2.17	102[c]	3.65	1.06
MC22-2-23	1	0.324	2.22	0.299	7.35	5.32	4.24	6.53	1.8
MC22-2-24	1	0.469	2.62	0.316	7.75	5.72	6.36	8.64	2.33
MC22-2-27	1	0.282	2.27	0.262	5.88	4.44	6.27	6.37	1.66
MC22-2-28	1	0.323	1.7	0.279	8.35	6.35	4.59	8.83	2.33
MC22-2-29	2	0.465	2.62	0.408	6.84	4.44	65.8[c]	5.06	1.31
MC22-2-30	1	0.222	2.34	0.276	5.74	3.27	212[c]	3.59	0.923
MC22-2-31	1	0.548	2.6	0.342	4.67	2.03	5.17	0.729	0.379
MC22-2-32	1	0.397	2.07	0.374	4.98	4.31	2.61	5.81	1.35
MC22-2-35	1	0.352	2.34	0.236	6.13	3.97	75.2[c]	4.62	1.02
MC22-2-40	1	0.368	2.01	0.253	6.87	4.16	19	4.98	1.38
MC22-2-5	1	0.289	2.11	0.263	6.33	4.83	3.76	6.57	1.62
MC22-2-8	1	0.211	1.79	0.153	3.98	3.61	155[c]	6	1.49
MC22-2-9	1	0.305	2.2	0.25	7.2	5.63	3.82	7.16	2
MC59-1-2	1	0.521	3.57	0.421	8.15	8.76	10.5	11.3	3.01
MC59-1-3	2	0.528	2.85	0.339	7.96	6.5	13.6	7.71	2.06
MC59-1-7	1	0.441	2.74	0.355	10.1	4.06	4.1	5.1	1.42
MC59-2-1	1	0.411	2.33	0.305	5.4	2.59	2.82	2.41	0.891
MC59-2-6	1	0.463	3.53	0.508	10.8	16.9	11.7	22.9	5.82
MC59-2-8	1	0.329	2.44	0.249	5.63	3.8	6.24	4.42	1.25
MC59-9-1	3	0.379	2.27	0.32	9.82	6.88	4.93	10.3	2.82
MC82-11	1	0.465	3.11	0.407	8.3	6.16	5.54	9.61	2.1
MC82-12	2	0.326	2.08	0.311	6.08	4.16	4.12	6.08	1.47
MC82-19	1	0.328	2.57	0.291	5.33	4.4	4.34	7.54	1.49
MC82-21	1	0.301	1.99	0.227	4.66	3.1	55.8[c]	4.53	1.3
MC82-28	1	0.76	4.04	0.289	8.26	11.2	1858[c]	9.36	2.1
MC82-29	1	0.346	2.5	0.357	6.54	4.32	4.3	6.06	1.41
MC82-32	1	0.207	1.74	0.224	3.35	2.45	3.11	5.07	0.913
MC82-35	1	0.476	2.86	0.379	7.96	7.04	6.97	9.41	2.53
MC82-37	1	0.351	2.44	0.327	7.01	4.41	7.22	5.62	1.37
MC82-39	1	0.344	2.19	0.279	6.38	4.25	4.48	5.57	1.54
MC82-40	1	0.42	2.02	0.319	6.21	4.71	4.12	5.93	1.44
MC82-5	1	0.431	2.53	0.331	7.54	7.18	1724[c]	9.92	2.5
MC82-6	1	0.389	2.1	0.276	5.57	3.8	3.83	6.06	1.27
MC99U-10	1	0.401	2.06	0.34	6.93	4.4	7.41	6.29	1.53
MC99U-7	1	0.345	2.12	0.327	9.6	8.1	7.76	12.7	2.87
BCR-2G measured	3	0.513 ± 0.020	3.57 ± 0.18	0.500 ± 0.043	4.88 ± 0.10	0.824 ± 0.048	11.04 ± 0.68	6.08 ± 0.11	1.84 ± 0.14
BCR-2G reference		0.510 ± 0.040	3.390 ± 0.030	0.5030 ± 0.0050	4.84 ± 0.28	0.780 ± 0.060	11.0 ± 1.0	5.90 ± 0.30	1.69 ± 0.12
BCR-2G accuracy (%)		+0.5	+5.3	−0.6	+0.9	+5.6	+0.3	+3.1	+8.6
KL2 test split measured	5	0.328 ± 0.028	2.07 ± 0.26	0.275 ± 0.031	3.88 ± 0.53	0.97 ± 0.16	2.090 ± 0.098	1.050 ± 0.090	0.57 ± 0.10
KL2 test split reference		0.3310 ± 0.0090	2.100 ± 0.050	0.2850 ± 0.0090	3.93 ± 0.14	0.961 ± 0.022	2.07 ± 0.10	1.020 ± 0.030	0.548 ± 0.016
KL2 test split accuracy (%)		−0.9	−1.4	−3.6	−1.3	+1.0	+1.0	+2.9	+4.3
KL2 precision, RSTD (%)		13.5	8.9	13.0	12.7	13.2	8.0	7.8	16.9

Concentrations in μg/g; repeat analyses of one inclusion averaged (n). Drift corrected with a linear fit of CF = measured / reference of the KL2 train analyses against time: corrected = raw / CF(time); all elements except ^{43}Ca.
* flagged element (step 6): RSTD of KL2 above 20 % or BCR-2G accuracy worse than ±15 %.
c: probable Pb contamination: Pb 56 to 1858 μg/g, against 2.6 to 22 μg/g in the other analyses; reported but not interpreted.
h: in the inclusions, partly host olivine ablated with the inclusion: the major-element oxide sums range from 67 to 416 wt% and correlate with these elements (r = 0.69 to 0.99); not interpreted.
^{43}Ca (IS): internal standard, fixed in GLITTER to a value entered for each analysis; not measured or corrected.
BCR-2G: run 70 excluded (normalised to the wrong Ca in GLITTER). KL2 test split: runs 6, 26, 46, 67, 83, not used for the correction. KL2 precision: train and test analyses together.
Reference values: GeoReM (Standards.xlsx) ± 95 % uncertainty; CI chondrite and primitive mantle: Sun and McDonough (1989).

Table A4: LA-ICP-MS standards measured as unknowns: accuracy and precision (RSTD) per element, in %

	KL2-G (n = 5 / 16)		BCR-2G (n = 3)		GOR132-G (n = 4)		GOR128-G (n = 4)		BM90/21-G (n = 2)	
	accuracy	RSTD	accuracy	RSTD	accuracy	RSTD	accuracy	RSTD	accuracy	RSTD
^{7}Li	+0.9	11.3	−13.7	3.9	+2.8	1.4	−1.6	17.5	−12.7	22.7
^{11}B *	+4.7	26.6	−30.5	30.4	−6.8	28.1	−14.3	22.2	−48.5	6.8
^{23}Na	+2.0	10.4	−2.7	5.0	+6.3	6.0	+12.1	17.8	+3.7	0.6
^{24}Mg	+2.0	8.6	−1.8	4.2	+5.9	4.6	+7.2	34.2	+2.0	0.1
^{27}Al	+1.3	7.2	+1.1	2.0	−1.3	4.3	+1.7	8.7	−0.7	5.1
^{29}Si	−1.2	5.5	−4.7	4.5	−1.3	5.8	+0.5	19.4	−3.2	0.1
^{31}P *	−2.5	11.7	−19.5	8.4	−26.9	12.1	−23.9	16.2	+910	4.6
^{39}K	−0.4	5.1	−0.2	3.9	−32.0	13.5	−24.4	19.8		
^{45}Sc	−0.9	4.5	+5.0	3.2	−3.6	7.5	−2.9	9.0	+24.6	0.6
^{47}Ti	−0.1	6.3	+1.0	0.3	−2.1	2.7	+0.3	4.0	−2.2	7.4
^{51}V	+1.8	9.0	−1.6	3.5	+8.0	5.6	+1.7	7.8	−5.4	0.3
^{55}Mn	+2.4	10.4	+2.9	6.3	+6.1	3.5	+7.3	10.3	+2.9	1.3
^{57}Fe	+3.2	9.8	−5.8	7.2	+8.3	5.4	+7.9	12.2	+9.0	1.2
^{75}As *	+4.5	48.0		9.7	−3.2	33.6	+59.0	94.2	+299	
^{85}Rb	+1.2	7.7	−0.4	2.1	+3.9	10.5	−0.9	31.6	−23.7	21.0
^{88}Sr	+2.5	5.7	−1.1	1.6	+1.4	4.6	+1.9	8.1	−14.4	4.1
^{89}Y	+0.5	5.0	+0.8	3.3	−0.4	9.7	−3.7	7.5	+2.4	6.9
^{90}Zr	+0.2	4.1	−0.3	1.3	−2.4	7.9	−4.8	8.7	−0.5	3.6
^{93}Nb	+0.9	6.7	+1.6	5.1	−35.8	77.1	+3.3	35.9	−42.0	
^{133}Cs *	+7.5	26.9	−4.4	10.3	+5.1	4.1	−13.3	15.2	−22.8	10.0
^{138}Ba	+1.5	11.6	−0.5	2.0	−3.9	22.3	+1.7	19.8	−13.4	7.2
^{139}La	+1.3	11.7	−2.3	2.4	+5.8	26.3	−8.9	11.2	−12.1	27.4
^{140}Ce	+0.8	7.3	−3.2	2.3	−12.8	10.6	−1.5	10.8	−7.5	9.5
^{141}Pr	+0.7	4.8	−4.2	1.7	−3.4	3.5	−12.4	22.2	+1.1	9.3
^{146}Nd	−1.1	7.8	−1.5	4.5	−9.9	30.1	−8.8	11.1	+10.8	8.2
^{147}Sm	+1.6	8.5	−8.2	7.7	−0.3	10.7	−3.7	18.5	−12.2	48.3
^{151}Eu	+1.0	11.7	+0.9	1.5	+5.5	10.5	+2.0	11.5	−5.8	26.4
^{157}Gd	+4.0	10.5	+1.0	7.2	+0.5	12.6	+3.9	19.7	+14.1	1.0
^{159}Tb	+3.4	8.6	−0.3	3.1	−8.6	21.7	−5.1	6.9	−12.8	4.8
^{163}Dy	−1.0	9.3	−2.5	5.3	−3.2	7.6	−5.8	6.1	+1.3	10.7
^{165}Ho	−2.0	12.6	+1.8	6.4	−4.5	12.6	−4.2	16.4	+3.6	11.4
^{166}Er	+3.3	14.0	+1.4	3.6	+0.7	12.9	−2.0	6.2	−10.1	25.1
^{169}Tm	−0.9	13.5	+0.5	3.9	−5.2	6.9	−8.3	30.8	+4.2	12.8
^{172}Yb	−1.4	8.9	+5.3	5.0	−9.2	7.9	−8.9	16.7	+10.3	7.8
^{175}Lu	−3.6	13.0	−0.6	8.5	−6.3	14.7	−6.5	31.5	+19.6	3.3
^{178}Hf	−1.3	12.7	+0.9	2.1	−1.6	16.5	+4.8	23.1	+7.1	11.7
^{181}Ta	+1.0	13.2	+5.6	5.8	−18.3	66.7	−7.9	6.7	+50.8	
^{208}Pb	+1.0	8.0	+0.3	6.2	+13.2	3.3	−10.4	3.2	−15.3	1.2
^{232}Th	+2.9	7.8	+3.1	1.9	−17.6		+55.3		+1.1	3.8
^{238}U	+4.3	16.9	+8.6	7.8	+24.8	29.1	−9.7	27.4	+3.4	23.7

Accuracy: deviation of the mean from the reference value; RSTD: 2SD / mean (precision); both in %, after step 6 (values below the detection limit and BCR-2G run 70 excluded). Blank: no reference value, no value above the detection limit, or fewer than two values for the RSTD.
KL2-G: accuracy from the test split (n = 5; the train analyses built the correction), RSTD from train and test together (n = 16).
GOR132-G and GOR128-G (komatiite glasses) and BM90/21-G (peridotite glass) are poor in incompatible elements (La 0.09 to 0.20 μg/g): values below the detection limit reduce n for As, Nb, Ta, Th and U, and the median deviation is 5.3, 5.5 and 10.1 %, against 1.8 % for BCR-2G and 1.4 % for KL2-G.
* flagged element (step 6). ^{43}Ca, the internal standard, is not shown. Reference values: GeoReM (Standards.xlsx).